

AYMANE AIT DADS

AI Software Intern | Agentic AI · LLM Fine-Tuning · Intelligent Automation

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and owning an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer - directly demonstrating the LLM orchestration and agent-based reasoning UL Solutions requires. Reduced manual network-analysis time by 60% at Orange Maroc by building automated ML pipelines on 100K+ records, translating model outputs into actionable maintenance recommendations for enterprise stakeholders. Brings hands-on experience in agentic pipelines, LLM fine-tuning, RAG, tool-calling, and intelligent automation across complex workflows, ready to prototype and evaluate agent-based AI solutions with an experienced engineering team. Aims to apply ablation-driven experimentation and chain-of-thought reasoning techniques to UL Solutions' enterprise AI platform, accelerating the development of robust, production-ready agentic systems.

CORE COMPETENCIES

Agentic AI Systems | Large Language Models | Intelligent Automation | LLM Orchestration | Prototype & Evaluate | Complex Workflow Reasoning | Enterprise Software Platforms | Agent-Based AI Solutions

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built automated ML pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records**, classifying network quality across thousands of nodes for the network optimization team.
- Developed end-to-end clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly into production network maintenance workflows.
- Reduced manual analysis time by **60%** through automated feature engineering, model evaluation, and intelligent automation of previously manual diagnostic processes.
- Delivered stakeholder presentations translating **complex model outputs** into actionable maintenance recommendations, bridging data science findings with enterprise business decisions.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in an ongoing NVIDIA reasoning challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context, all attention + MLP + lm_head projections targeted).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers generating 64+ verified chain-of-thought rows.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · 1st in Class

- Ranked **1st in EURECOM class** private Kaggle leaderboard (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning-rate schedule on 250K images spanning 25+ generator types.
- Submitted independently to **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC), validating early-stop strategy for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM & Agentic AI: LLM Fine-Tuning (LoRA, QLoRA) | Agentic Pipelines | LangChain | RAG | Tool-Calling | Prompt Engineering | vLLM

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Unsloth · TRL · PEFT | Scikit-learn | BF16/FP16 Mixed Precision | Ablation Studies

Data, Cloud & Infra: Python | Pandas · NumPy | SQL | REST APIs | Supabase | Docker (basic) | Git | Power BI